MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : ES-EE101/ESEE101 Basic Electrical Engineering

UPID : 001009

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.
Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(i) What is the formula for output voltage for Buck converter?

(ii) What is the full form of MCCB?

(iii) Draw the Norton's equivalent circuit

(iv) What is the Power factor of electric incandescent bulb?

(v) What is the condition for maximum efficiency of transformer?

(vi) Explain the law based on which a DC generator works.

(vii) A single-phase full bridge inverter has a dc voltage source Vs = 230 V. Find the rms value of the fundamental component of output voltage.

(viii) How the capacity of a battery is expressed?

(ix) What is meant by short and open circuits?

(x) The r.m.s. value of a sinusoidal A.C. current is equal to its value at an angle of _____ degrees.

(xi) Why SC test is generally performed on HV side of a transformer?

(xii) What is mechanical rectifier related to DC machine?

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. What is meant by power converter? [5]

3. Explain what is meant by series resonance of a series RLC ac circuit. How do you define of such circuit? [5]

4. Calculate the resistance of 100 m length of a wire having a uniform cross-sectional area of 0.1 mm^2 if the wire is made of manganin having a resistivity of 50×10^{-8} Ω-m. [5]

5. Show that the power consumed in a pure inductance is zero. [5]

6. A. d.c. Shunt generator supplies a load of 50KW at 500 V. Calculate the emf generated if the armature and shunt field resistances are 0.25 ohm and 125 ohm respectively. [5]

Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. (a) What is MCCB and how does it differ from MCB? [4]

(b) What is earthing? Why earthing is provided? [4]

(c) An alkaline cell is discharged at a steady current of 4 A for 12 hours, the average terminal voltage being 1.2 V. To restore to its original state of charge, a steady current of 3 A for 20 hours is required, the average terminal voltage being 1.44 V. Calculate watt-hour efficiency. [7]

8. (a) Draw the phasor diagram of a transformer on load and in lagging power factor [7]

(b) The maximum efficiency of a 100 kVA, single phase transformer is 95% and occurs at 90% of full load at 0.85 pf. If the leakage impedance of the transformer if 5%, find the voltage regulation at rated load 0.8 pf lagging. [8]

9. (a) Define and explain the following: (i) time period (ii) amplitude (iii) phase difference. [6]

(b) A 50-μF capacitor is connected across a 230-V, 50-Hz supply. Calculate (a) the reactance offered by the capacitor (b) the maximum current and (c) the r.m.s. value of the current drawn by the capacitor. [9]

10. (a) Derive the EMF equation of a transformer. [4]

(b) A 8 kVA, 440/2000 V, 50 Hz single phase transformer gave the following test results: [8]
No load test: 400 V, 0.8 A, 80 W

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Short circuit test: 50 V, 3 A, 20W

Calculate (i) the magnetizing current and the component corresponding to Iron losses at normal voltage and frequency, (ii) the efficiency on full load at unity power factor, (iii) the secondary terminal voltage on full load at unity power factor. [3]

(c) Draw the equivalent circuit of a transformer refer to the primary side. [8]

11. (a) A discharged battery is put on charge at 5 A for 3.5 hours at a mean charging voltage of 13.5 V. It is then discharged in 6 hours at constant terminal voltage of 12 V through a resistance of R ohms. Calculate the value of R for an A-H efficiency of 85%. Also calculate the watt-hour efficiency. [7]

(b) A battery is charged keeping the voltage constant. The current initially was 10.5 A and finally 2.5 A in a period of 8 hours. The battery was discharged at a constant current of 6 A for 7 hours. Calculate ampere-hour efficiency of this battery.

*** END OF PAPER ***

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